Mini project report on

Blockchain Transaction Explorer & Analytics Platform

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering

UE23CS351A – DBMS Project

Submitted by:

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under the guidance of

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Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

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CERTIFICATE

This is to certify that the mini project entitled

Disasters data Management System

is a bonafide work carried out by

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In partial fulfilment for the completion of fifth semester DBMS Project (UE24CSS351A) in the Program of Study -Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period AUG. 2026 – DEC. 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 5th semester academic requirements in respect of project work.

Signature

Dr. Shilpa S

Assistant Professor

DECLARATION

We hereby declare that the DBMS Project entitled **Disasters Data Management System** has been carried out by us under the guidance of **Dr. Shilpa S, Assistant Professor** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester AUG – DEC 2026.

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ABSTRACT

The Blockchain Transaction Explorer & Analytics Platform is a database management system designed to organize, manage and analyse blockchain-related information, including blocks, transactions, wallet addresses and token transfers.

The system uses a hybrid database architecture consisting of MySQL and MongoDB. MySQL manages structured relational information through seven main tables: users, watchlist, watchlist_address, address, block, blockchain_transaction and token_transfer. It uses primary keys, foreign keys, unique constraints and check constraints to maintain data integrity and consistency.

MongoDB is integrated as a NoSQL database to support two additional functionalities: Wallet Activity History and Smart Contract Event Explorer. The Wallet Activity History feature maintains detailed information about wallet interactions, including incoming and outgoing transactions. The Smart Contract Event Explorer stores flexible event information, such as transaction hashes, contract addresses, event names and event-specific data.

The project demonstrates important Database Management System concepts, including Data Definition Language (DDL), Data Manipulation Language (DML), SQL queries, joins, aggregate functions, views, stored procedures, triggers and MongoDB CRUD operations.

The platform provides an integrated interface for exploring blockchain transactions, monitoring wallet activities, examining smart contract events and analysing blockchain-related information. This project demonstrates the practical application of relational and NoSQL databases in developing a hybrid database management system.

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1. INTRODUCTION

Blockchain technology is a decentralized digital ledger that records transactions across a distributed network. Every transaction is stored in a block, and blocks are linked together to maintain the integrity and security of the blockchain.

A Blockchain Transaction Explorer is a system that allows users to examine blockchain blocks, transactions, wallet addresses and token transfers. Managing this information requires an organized database system that can efficiently store, retrieve and analyse large amounts of related data.

The Blockchain Transaction Explorer & Analytics Platform is a Database Management System project developed using MySQL and MongoDB. The system follows a hybrid database architecture in which MySQL is used to manage structured relational information, while MongoDB is used to store flexible and document-oriented blockchain information.

The MySQL database consists of seven main tables: users, watchlist, watchlist_address, address, block, blockchain_transaction and token_transfer. These tables maintain relationships between users, wallet addresses, blocks and transactions using primary keys, foreign keys and integrity constraints.

MongoDB is incorporated to provide additional functionalities through Wallet Activity History and Smart Contract Event Explorer. The Wallet Activity History feature maintains detailed information about incoming and outgoing transactions associated with wallet addresses. The Smart Contract Event Explorer stores and displays smart contract event information, including event names, contract addresses, transaction hashes and event-specific data.

The project demonstrates important database concepts such as relational database design, primary and foreign keys, integrity constraints, SQL queries, joins, aggregate functions, views, stored procedures, triggers and MongoDB CRUD operations.

The primary objective of this project is to develop an integrated database management system that combines relational and NoSQL database technologies to efficiently manage, retrieve and analyse blockchain-related information.

2. PROBLEM DEFINITION

Blockchain networks generate large volumes of information, including transaction records, block details, wallet addresses, token transfers and smart contract events. Managing and analysing this information requires an organized database system capable of handling both structured and flexible data.

A traditional relational database can efficiently manage structured blockchain information and maintain relationships between different entities. However, additional information such as detailed wallet activity records and smart contract event data may contain varying attributes and nested structures.

The objective of this project is to develop a hybrid database management system using MySQL and MongoDB to efficiently store, retrieve and analyse blockchain-related information.

Objectives:

1. Store blockchain block information and transaction records.
2. Maintain wallet addresses and their balances.
3. Track token transfers between wallet addresses.
4. Allow users to create and manage watchlists.
5. Maintain relationships between blocks, transactions and addresses.
6. Enforce data integrity using primary keys, foreign keys and constraints.
7. Retrieve transaction information using SQL queries and joins.
8. Analyse transaction data using aggregate functions and grouping.
9. Implement stored procedures for retrieving transactions.
10. Use triggers to validate transaction data before insertion.
11. Integrate MongoDB to store flexible and document-oriented blockchain information.
12. Maintain detailed wallet activity history, including incoming and outgoing transactions.
13. Implement a Smart Contract Event Explorer to store, retrieve and examine smart contract event information.
14. Perform MongoDB CRUD operations to manage wallet activity and smart contract event records.

3. ER MODEL

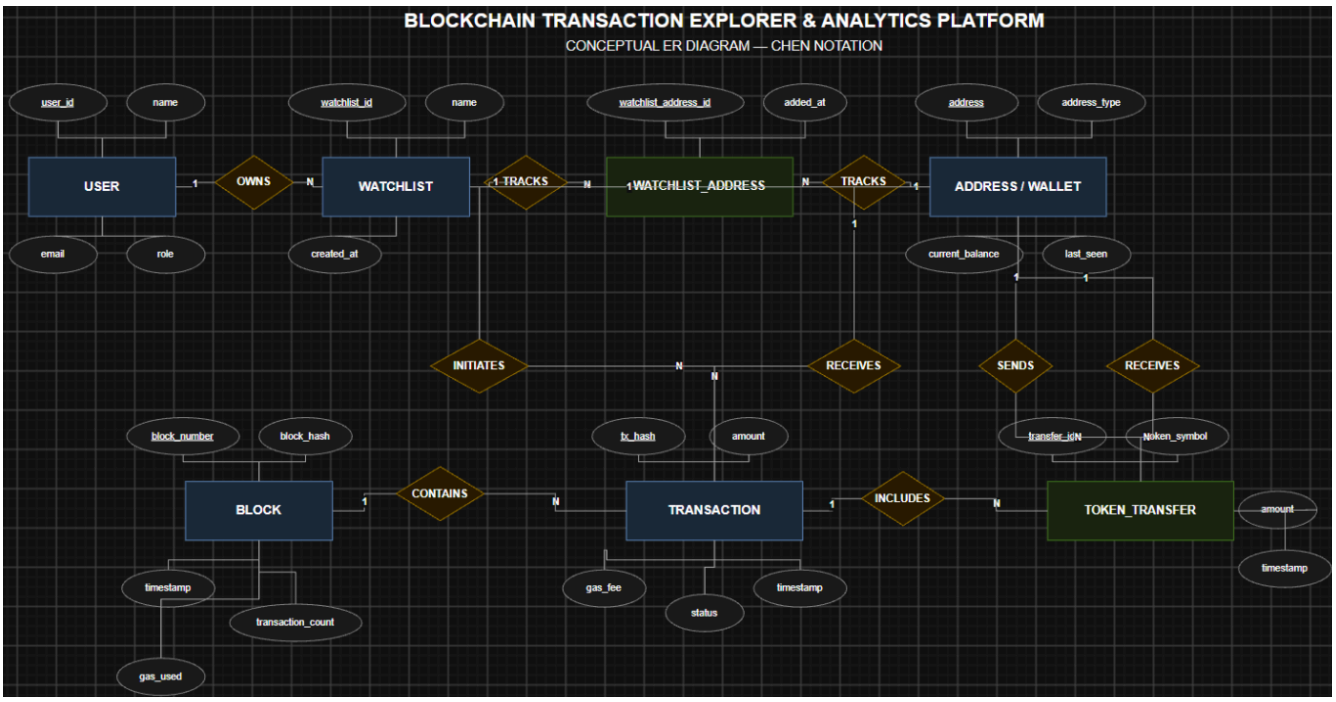


Figure 3.1: Entity Relationship Diagram of Blockchain Transaction Explorer & Analytics Platform

The ER diagram represents the entities and relationships involved in the Blockchain Transaction Explorer & Analytics Platform.

Entities:

1. Users – Stores user information such as user ID, name, email and role.
2. Watchlist – Stores watchlists created by users.
3. Watchlist_Address – Maintains the relationship between watchlists and wallet addresses.
4. Address – Stores wallet addresses, address types, current balances and last-seen timestamps.
5. Block – Stores blockchain block information, including block number, block hash and gas usage.
6. Blockchain_Transaction – Stores transaction hashes, amounts, gas fees, status and associated block and wallet addresses.
7. Token_Transfer – Stores token transfer details, including token symbol, amount and associated transaction.

The relationships ensure that users can maintain watchlists, addresses can participate in transactions, blocks contain transactions, and transactions can have associated token transfers.

4. ER TO RELATIONAL MAPPING

4.1 STEPS OF ALGORITHM FOR CHOSEN PROBLEM

The ER model is converted into a relational database schema using the following steps:

1. Identify all entities in the ER diagram.
2. Create a separate relation for each entity.
3. Identify the attributes associated with each entity.
4. Select primary keys to uniquely identify each record.
5. Map one-to-many relationships by including the primary key of the referenced relation as a foreign key.
6. Map many-to-many relationships using an intermediate relation.
7. Apply appropriate data types and integrity constraints.
8. Implement the resulting relational schema using MySQL.

The resulting database consists of seven relations: users, watchlist, watchlist_address, address, block, blockchain_transaction and token_transfer.

4.2 COMPLETE DIAGRAM OF RELATIONAL MAPPING

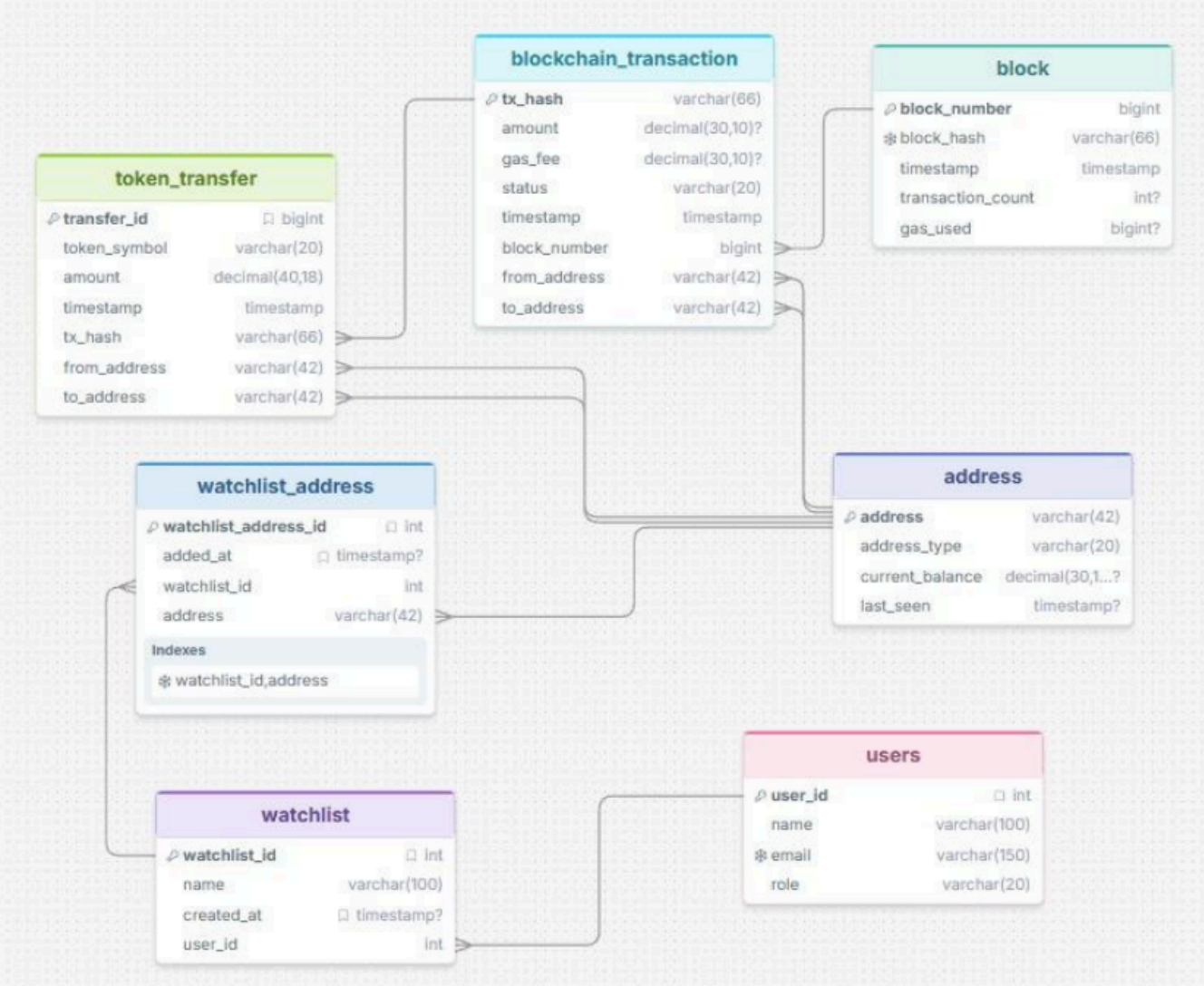


Figure 4.1: Relational Schema Diagram

5. DDL STATEMENTS

5.1 Database creation

```
mysql> CREATE DATABASE blockchain_explorer;
Query OK, 1 row affected (0.03 sec)

mysql> use blockchain_explorer;
Database changed
```

5.2 Users table

```
mysql> CREATE TABLE users (
    ->     user_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(100) NOT NULL,
    ->     email VARCHAR(150) NOT NULL UNIQUE,
    ->     role VARCHAR(20) NOT NULL
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql> desc users;
+---------+--------------+------+-----+---------+----------------+
| Field   | Type         | Null | Key | Default | Extra          |
+---------+--------------+------+-----+---------+----------------+
| user_id | int          | NO   | PRI | NULL    | auto_increment |
| name    | varchar(100) | NO   |     | NULL    |                |
| email   | varchar(150) | NO   | UNI | NULL    |                |
| role    | varchar(20)  | NO   |     | NULL    |                |
+---------+--------------+------+-----+---------+----------------+
4 rows in set (0.02 sec)
```

5.3 Watchlist table

```
mysql> CREATE TABLE watchlist (
    ->     watchlist_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     name VARCHAR(100) NOT NULL,
    ->     created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     user_id INT NOT NULL,
    ->
    ->     FOREIGN KEY (user_id)
    ->         REFERENCES users(user_id)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> desc watchlist;
+--------------+--------------+------+-----+-------------------+-------------------+
| Field        | Type         | Null | Key | Default           | Extra             |
+--------------+--------------+------+-----+-------------------+-------------------+
| watchlist_id | int          | NO   | PRI | NULL              | auto_increment    |
| name         | varchar(100) | NO   |     | NULL              |                   |
| created_at   | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| user_id      | int          | NO   | MUL | NULL              |                   |
+--------------+--------------+------+-----+-------------------+-------------------+
4 rows in set (0.00 sec)
```

5.4 Address table

```
mysql> CREATE TABLE address (
    ->     address VARCHAR(42) PRIMARY KEY,
    ->     address_type VARCHAR(20) NOT NULL,
    ->     current_balance DECIMAL(30,10) DEFAULT 0,
    ->     last_seen TIMESTAMP NULL
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> desc address;
+-----------------+---------------+------+-----+--------------+-------+
| Field           | Type          | Null | Key | Default      | Extra |
+-----------------+---------------+------+-----+--------------+-------+
| address         | varchar(42)   | NO   | PRI | NULL         |       |
| address_type    | varchar(20)   | NO   |     | NULL         |       |
| current_balance | decimal(30,10)| YES  |     | 0.0000000000 |       |
| last_seen       | timestamp     | YES  |     | NULL         |       |
+-----------------+---------------+------+-----+--------------+-------+
4 rows in set (0.00 sec)
```

5.5 Watchlist_Address table

```
mysql> CREATE TABLE watchlist_address (
    ->     watchlist_address_id INT PRIMARY KEY AUTO_INCREMENT,
    ->     added_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    ->     watchlist_id INT NOT NULL,
    ->     address VARCHAR(42) NOT NULL,
    ->
    ->     FOREIGN KEY (watchlist_id)
    ->         REFERENCES watchlist(watchlist_id),
    ->
    ->     FOREIGN KEY (address)
    ->         REFERENCES address(address),
    ->
    ->     UNIQUE (watchlist_id, address)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> desc watchlist_address;
+---------------------+-------------+------+-----+-------------------+-------------------+
| Field               | Type        | Null | Key | Default           | Extra             |
+---------------------+-------------+------+-----+-------------------+-------------------+
| watchlist_address_id | int         | NO   | PRI | NULL              | auto_increment    |
| added_at            | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| watchlist_id        | int         | NO   | MUL | NULL              |                   |
| address             | varchar(42) | NO   | MUL | NULL              |                   |
+---------------------+-------------+------+-----+-------------------+-------------------+
4 rows in set (0.00 sec)
```

5.6 Block table

```
mysql> CREATE TABLE block (
    ->     block_number BIGINT PRIMARY KEY,
    ->     block_hash VARCHAR(66) NOT NULL UNIQUE,
    ->     timestamp TIMESTAMP NOT NULL,
    ->     transaction_count INT DEFAULT 0,
    ->     gas_used BIGINT DEFAULT 0
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> desc block;
+-------------------+-------------+------+-----+---------+-------+
| Field             | Type        | Null | Key | Default | Extra |
+-------------------+-------------+------+-----+---------+-------+
| block_number      | bigint      | NO   | PRI | NULL    |       |
| block_hash        | varchar(66) | NO   | UNI | NULL    |       |
| timestamp         | timestamp   | NO   |     | NULL    |       |
| transaction_count | int         | YES  |     | 0       |       |
| gas_used          | bigint      | YES  |     | 0       |       |
+-------------------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

5.7 Blockchain_Transaction table

```
mysql> CREATE TABLE blockchain_transaction (
    ->     tx_hash VARCHAR(66) PRIMARY KEY,
    ->     amount DECIMAL(30,10) DEFAULT 0,
    ->     gas_fee DECIMAL(30,10) DEFAULT 0,
    ->     status VARCHAR(20) NOT NULL,
    ->     timestamp TIMESTAMP NOT NULL,
    ->
    ->     block_number BIGINT NOT NULL,
    ->     from_address VARCHAR(42) NOT NULL,
    ->     to_address VARCHAR(42) NOT NULL,
    ->
    ->     FOREIGN KEY (block_number)
    ->         REFERENCES block(block_number),
    ->
    ->     FOREIGN KEY (from_address)
    ->         REFERENCES address(address),
    ->
    ->     FOREIGN KEY (to_address)
    ->         REFERENCES address(address)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> desc blockchain_transaction;
+--------------+----------------+------+-----+-------------+-------+
| Field        | Type           | Null | Key | Default     | Extra |
+--------------+----------------+------+-----+-------------+-------+
| tx_hash      | varchar(66)    | NO   | PRI | NULL        |       |
| amount       | decimal(30,10) | YES  |     | 0.0000000000 |       |
| gas_fee      | decimal(30,10) | YES  |     | 0.0000000000 |       |
| status       | varchar(20)    | NO   |     | NULL        |       |
| timestamp    | timestamp      | NO   |     | NULL        |       |
| block_number | bigint         | NO   | MUL | NULL        |       |
| from_address | varchar(42)    | NO   | MUL | NULL        |       |
| to_address   | varchar(42)    | NO   | MUL | NULL        |       |
+--------------+----------------+------+-----+-------------+-------+
8 rows in set (0.00 sec)
```

5.8 Token_Transfer table

```
mysql> CREATE TABLE token_transfer (
    ->     transfer_id BIGINT PRIMARY KEY AUTO_INCREMENT,
    ->     token_symbol VARCHAR(20) NOT NULL,
    ->     amount DECIMAL(40,18) NOT NULL,
    ->     timestamp TIMESTAMP NOT NULL,
    ->
    ->     tx_hash VARCHAR(66) NOT NULL,
    ->     from_address VARCHAR(42) NOT NULL,
    ->     to_address VARCHAR(42) NOT NULL,
    ->
    ->     FOREIGN KEY (tx_hash)
    ->         REFERENCES blockchain_transaction(tx_hash),
    ->
    ->     FOREIGN KEY (from_address)
    ->         REFERENCES address(address),
    ->
    ->     FOREIGN KEY (to_address)
    ->         REFERENCES address(address)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> desc token_transfer;
+--------------+----------------+------+-----+---------+----------------+
| Field        | Type           | Null | Key | Default | Extra          |
+--------------+----------------+------+-----+---------+----------------+
| transfer_id  | bigint         | NO   | PRI | NULL    | auto_increment |
| token_symbol | varchar(20)    | NO   |     | NULL    |                |
| amount       | decimal(40,18) | NO   |     | NULL    |                |
| timestamp    | timestamp      | NO   |     | NULL    |                |
| tx_hash      | varchar(66)    | NO   | MUL | NULL    |                |
| from_address | varchar(42)    | NO   | MUL | NULL    |                |
| to_address   | varchar(42)    | NO   | MUL | NULL    |                |
+--------------+----------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

6. DML STATEMENTS

6.1 INSERT - Users

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO users (name, email, role) VALUES
    -> ('Arjun Sharma', 'arjun.sharma@gmail.com', 'Admin'),
    -> ('Rahul Kumar', 'rahul.kumar@gmail.com', 'Analyst'),
    -> ('Sneha Rao', 'sneha.rao@gmail.com', 'User'),
    -> ('Aditya Singh', 'aditya.singh@gmail.com', 'User'),
    -> ('Priya Nair', 'priya.nair@gmail.com', 'Analyst'),
    -> ('Karan Patel', 'karan.patel@gmail.com', 'User'),
    -> ('Ananya Das', 'ananya.das@gmail.com', 'User'),
    -> ('Rohit Verma', 'rohit.verma@gmail.com', 'Analyst'),
    -> ('Meera Iyer', 'meera.iyer@gmail.com', 'User'),
    -> ('Vikram Joshi', 'vikram.joshi@gmail.com', 'Admin');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

6.2 INSERT - Address

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO address
    -> (address, address_type, current_balance, last_seen)
    -> VALUES
    -> ('0x1111111111111111111111111111111111111111',
    ->  'EOA', 12.5000000000, '2026-09-01 10:00:00'),
    ->
    -> ('0x2222222222222222222222222222222222222222',
    ->  'EOA', 25.7500000000, '2026-09-02 11:00:00'),
    ->
    -> ('0x3333333333333333333333333333333333333333',
    ->  'Contract', 100.0000000000, '2026-09-03 12:00:00'),
    ->
    -> ('0x4444444444444444444444444444444444444444',
    ->  'EOA', 8.2500000000, '2026-09-04 13:00:00'),
    ->
    -> ('0x5555555555555555555555555555555555555555',
    ->  'EOA', 50.0000000000, '2026-09-05 14:00:00'),
    ->
    -> ('0x6666666666666666666666666666666666666666',
    ->  'Contract', 75.5000000000, '2026-09-06 15:00:00'),
    ->
    -> ('0x7777777777777777777777777777777777777777',
    ->  'EOA', 18.7500000000, '2026-09-07 16:00:00'),
    ->
    -> ('0x8888888888888888888888888888888888888888',
    ->  'EOA', 32.0000000000, '2026-09-08 17:00:00'),
    ->
    -> ('0x9999999999999999999999999999999999999999',
    ->  'Contract', 200.0000000000, '2026-09-09 18:00:00'),
    ->
    -> ('0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa',
    ->  'EOA', 5.5000000000, '2026-09-10 19:00:00'),
    ->
    -> ('0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
    ->  'EOA', 45.0000000000, '2026-09-11 10:00:00'),
    ->
    -> ('0xcccccccccccccccccccccccccccccccccccccccc',
    ->  'Contract', 150.0000000000, '2026-09-12 11:00:00');
Query OK, 12 rows affected (0.00 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

6.3 INSERT - Block

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO block
    -> (block_number, block_hash, timestamp, transaction_count, gas_used)
    -> VALUES
    -> (100001,
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa',
    ->  '2026-09-01 10:00:00', 2, 210000),
    ->
    -> (100002,
    ->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
    ->  '2026-09-02 11:00:00', 2, 280000),
    ->
    -> (100003,
    ->  '0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc',
    ->  '2026-09-03 12:00:00', 2, 150000),
    ->
    -> (100004,
    ->  '0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd',
    ->  '2026-09-04 13:00:00', 2, 350000),
    ->
    -> (100005,
    ->  '0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee',
    ->  '2026-09-05 14:00:00', 2, 420000),
    ->
    -> (100006,
    ->  '0xffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff',
    ->  '2026-09-06 15:00:00', 2, 230000),
    ->
    -> (100007,
    ->  '0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef',
    ->  '2026-09-07 16:00:00', 2, 300000),
    ->
    -> (100008,
    ->  '0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcd',
    ->  '2026-09-08 17:00:00', 2, 170000),
    ->
    -> (100009,
    ->  '0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdef',
    ->  '2026-09-09 18:00:00', 2, 390000),
    ->
    -> (100010,
    ->  '0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc',
    ->  '2026-09-10 19:00:00', 2, 250000);
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

6.4 INSERT - Watchlist

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO watchlist
    -> (name, user_id)
    -> VALUES
    -> ('High Value Wallets', 1),
    -> ('DeFi Addresses', 2),
    -> ('Whale Tracking', 3),
    -> ('Exchange Wallets', 4),
    -> ('Smart Contracts', 5),
    -> ('Suspicious Addresses', 6),
    -> ('Token Monitoring', 7),
    -> ('Daily Transactions', 8),
    -> ('Large Transfers', 9),
    -> ('Research Wallets', 10);
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

6.5 INSERT - Watchlist_Address

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO watchlist_address
    -> (watchlist_id, address)
    -> VALUES
    -> (1, '0x1111111111111111111111111111111111111111'),
    -> (1, '0x5555555555555555555555555555555555555555'),
    ->
    -> (2, '0x3333333333333333333333333333333333333333'),
    -> (2, '0x6666666666666666666666666666666666666666'),
    ->
    -> (3, '0x9999999999999999999999999999999999999999'),
    -> (3, '0xcccccccccccccccccccccccccccccccccccccccc'),
    ->
    -> (4, '0x2222222222222222222222222222222222222222'),
    -> (4, '0x8888888888888888888888888888888888888888'),
    ->
    -> (5, '0x3333333333333333333333333333333333333333'),
    -> (5, '0x9999999999999999999999999999999999999999'),
    ->
    -> (6, '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'),
    ->
    -> (7, '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb'),
    ->
    -> (8, '0x7777777777777777777777777777777777777777'),
    ->
    -> (9, '0x8888888888888888888888888888888888888888'),
    ->
    -> (10, '0x4444444444444444444444444444444444444444');
Query OK, 15 rows affected (0.00 sec)
Records: 15  Duplicates: 0  Warnings: 0
```

6.6 INSERT - Blockchain_Transaction

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO blockchain_transaction
    -> (tx_hash, amount, gas_fee, status, timestamp,
    ->  block_number, from_address, to_address)
    -> VALUES
    ->
    -> ('0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'
    ->  5.5000000000, 0.0021000000, 'Success',
    ->  '2026-09-01 10:05:00', 100001,
    ->  '0x1111111111111111111111111111111111111111',
    ->  '0x2222222222222222222222222222222222222222'),
    ->
    -> ('0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaab'
    ->  8.2500000000, 0.0025000000, 'Success',
    ->  '2026-09-01 10:08:00', 100001,
    ->  '0x2222222222222222222222222222222222222222',
    ->  '0x3333333333333333333333333333333333333333'),
    ->
    -> ('0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
    ->  10.2500000000, 0.0035000000, 'Success',
    ->  '2026-09-02 11:10:00', 100002,
    ->  '0x3333333333333333333333333333333333333333',
    ->  '0x4444444444444444444444444444444444444444'),
    ->
    -> ('0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbc',
    ->  2.7500000000, 0.0015000000, 'Failed',
    ->  '2026-09-02 11:15:00', 100002,
    ->  '0x4444444444444444444444444444444444444444',
    ->  '0x5555555555555555555555555555555555555555'),
    ->
    -> ('0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc',
    ->  20.0000000000, 0.0045000000, 'Success',
    ->  '2026-09-03 12:20:00', 100003,
    ->  '0x5555555555555555555555555555555555555555',
    ->  '0x6666666666666666666666666666666666666666'),
    ->
    -> ('0xccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccd',
    ->  3.7500000000, 0.0018000000, 'Pending',
    ->  '2026-09-03 12:25:00', 100003,
    ->  '0x6666666666666666666666666666666666666666',
    ->  '0x7777777777777777777777777777777777777777'),
    ->
```

```
->
-> ('0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddd',
->  7.2500000000, 0.0028000000, 'Pending',
->  '2026-09-04 13:30:00', 100004,
->  '0x7777777777777777777777777777777777777777',
->  '0x8888888888888888888888888888888888888888'),
->
-> ('0xddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddddde',
->  15.7500000000, 0.0032000000, 'Success',
->  '2026-09-04 13:35:00', 100004,
->  '0x8888888888888888888888888888888888888888',
->  '0x9999999999999999999999999999999999999999'),
->
-> ('0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee',
->  30.5000000000, 0.0050000000, 'Success',
->  '2026-09-05 14:40:00', 100005,
->  '0x9999999999999999999999999999999999999999',
->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'),
->
-> ('0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeef',
->  12.0000000000, 0.0025000000, 'Failed',
->  '2026-09-05 14:45:00', 100005,
->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa',
->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb'),
->
-> ('0xffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff
->  15.7500000000, 0.0032000000, 'Success',
->  '2026-09-06 15:50:00', 100006,
->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
->  '0xcccccccccccccccccccccccccccccccccccccccc'),
->
-> ('0xffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffe
->  4.2500000000, 0.0017000000, 'Success',
->  '2026-09-06 15:55:00', 100006,
->  '0xcccccccccccccccccccccccccccccccccccccccc',
->  '0x1111111111111111111111111111111111111111'),
```

```
-> ('0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef',
->  3.5000000000, 0.0012000000, 'Success',
->  '2026-09-07 16:00:00', 100007,
->  '0x1111111111111111111111111111111111111111',
->  '0x4444444444444444444444444444444444444444'),
->
-> ('0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdee',
->  18.0000000000, 0.0040000000, 'Success',
->  '2026-09-07 16:05:00', 100007,
->  '0x4444444444444444444444444444444444444444',
->  '0x5555555555555555555555555555555555555555'),
->
-> ('0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcd',
->  22.5000000000, 0.0045000000, 'Failed',
->  '2026-09-08 17:10:00', 100008,
->  '0x5555555555555555555555555555555555555555',
->  '0x6666666666666666666666666666666666666666'),
->
-> ('0xabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabcdefabce',
->  6.7500000000, 0.0022000000, 'Pending',
->  '2026-09-08 17:15:00', 100008,
->  '0x6666666666666666666666666666666666666666',
->  '0x7777777777777777777777777777777777777777'),
->
-> ('0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdef',
->  40.0000000000, 0.0060000000, 'Success',
->  '2026-09-09 18:20:00', 100009,
->  '0x7777777777777777777777777777777777777777',
->  '0x8888888888888888888888888888888888888888'),
->
-> ('0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdee',
->  9.5000000000, 0.0027000000, 'Success',
->  '2026-09-09 18:25:00', 100009,
->  '0x8888888888888888888888888888888888888888',
->  '0x9999999999999999999999999999999999999999'),
```

```
    -> ('0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc',
    ->  25.0000000000, 0.0048000000, 'Success',
    ->  '2026-09-10 19:30:00', 100010,
    ->  '0x9999999999999999999999999999999999999999',
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'),
    ->
    -> ('0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabd',
    ->  11.2500000000, 0.0031000000, 'Failed',
    ->  '2026-09-10 19:35:00', 100010,
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa',
    ->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb');
Query OK, 20 rows affected (0.01 sec)
Records: 20  Duplicates: 0  Warnings: 0
```

6.7 INSERT - Token_Transfer

```
PES2UG24CS646_PES2UG24CS916>INSERT INTO token_transfer
    -> (token_symbol, amount, timestamp, tx_hash,
    ->  from_address, to_address)
    -> VALUES
    ->
    -> ('USDT', 1000.00000000000000000,
    ->  '2026-09-01 10:05:30',
    ->  '0xaaa
```

```
    ->
    -> ('UNI', 300.00000000000000000,
    ->  '2026-09-05 14:40:30',
    ->  '0xeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeeee',
    ->  '0x999999999999999999999999999999999999999999',
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'),
    ->
    -> ('USDC', 1250.00000000000000000,
    ->  '2026-09-06 15:50:30',
    ->  '0xffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff',
    ->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb',
    ->  '0xcccccccccccccccccccccccccccccccccccccccc'),
    ->
    -> ('USDT', 4000.00000000000000000,
    ->  '2026-09-07 16:00:30',
    ->  '0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef',
    ->  '0x1111111111111111111111111111111111111111',
    ->  '0x4444444444444444444444444444444444444444'),
    ->
    -> ('DAI', 800.00000000000000000,
    ->  '2026-09-07 16:05:30',
    ->  '0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdee',
    ->  '0x4444444444444444444444444444444444444444',
    ->  '0x5555555555555555555555555555555555555555'),
    ->
    -> ('USDT', 3500.00000000000000000,
    ->  '2026-09-09 18:20:30',
    ->  '0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdef',
    ->  '0x7777777777777777777777777777777777777777',
    ->  '0x8888888888888888888888888888888888888888'),
    ->
    -> ('USDC', 600.00000000000000000,
    ->  '2026-09-09 18:25:30',
    ->  '0x9876543210abcdef9876543210abcdef9876543210abcdef9876543210abcdee',
    ->  '0x8888888888888888888888888888888888888888',
    ->  '0x9999999999999999999999999999999999999999'),
    ->
    -> ('ETH', 8.50000000000000000,
    ->  '2026-09-10 19:30:30',
    ->  '0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabc',
    ->  '0x9999999999999999999999999999999999999999',
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa'),
```

```
    -> ('LINK', 450.000000000000000000,
    ->  '2026-09-10 19:35:30',
    ->  '0xfedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbafedcbabd',
    ->  '0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa',
    ->  '0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb');
Query OK, 14 rows affected (0.00 sec)
Records: 14  Duplicates: 0  Warnings: 0
```

7. QUERIES

7.1 SIMPLE QUERY WITH GROUP BY, AGGREGATE

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     status,
    ->     COUNT(*) AS transaction_count
    -> FROM blockchain_transaction
    -> GROUP BY status;
+---------+-------------------+
| status  | transaction_count |
+---------+-------------------+
| Success |                13 |
| Failed  |                 4 |
| Pending |                 3 |
+---------+-------------------+
3 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     block_number,
    ->     COUNT(*) AS total_transactions,
    ->     SUM(amount) AS total_amount
    -> FROM blockchain_transaction
    -> GROUP BY block_number;
+--------------+--------------------+--------------+
| block_number | total_transactions | total_amount |
+--------------+--------------------+--------------+
|       100001 |                  2 | 13.7500000000 |
|       100002 |                  2 | 13.0000000000 |
|       100003 |                  2 | 23.7500000000 |
|       100004 |                  2 | 23.0000000000 |
|       100005 |                  2 | 42.5000000000 |
|       100006 |                  2 | 20.0000000000 |
|       100007 |                  2 | 21.5000000000 |
|       100008 |                  2 | 29.2500000000 |
|       100009 |                  2 | 49.5000000000 |
|       100010 |                  2 | 36.2500000000 |
+--------------+--------------------+--------------+
10 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     token_symbol,
    ->     COUNT(*) AS transfer_count,
    ->     SUM(amount) AS total_transferred,
    ->     AVG(amount) AS average_transfer
    -> FROM token_transfer
    -> GROUP BY token_symbol;
+--------------+----------------+-----------------------+---------------------------+
| token_symbol | transfer_count | total_transferred     | average_transfer          |
+--------------+----------------+-----------------------+---------------------------+
| USDT         |              4 | 11000.000000000000000000 | 2750.000000000000000000000 |
| USDC         |              3 |  2350.000000000000000000 |  783.333333333333333333 |
| DAI          |              2 |  1550.000000000000000000 |  775.000000000000000000000 |
| LINK         |              2 |   600.000000000000000000 |  300.000000000000000000000 |
| ETH          |              2 |    13.500000000000000000 |    6.750000000000000000000 |
| UNI          |              1 |   300.000000000000000000 |  300.000000000000000000000 |
+--------------+----------------+-----------------------+---------------------------+
6 rows in set (0.00 sec)
```

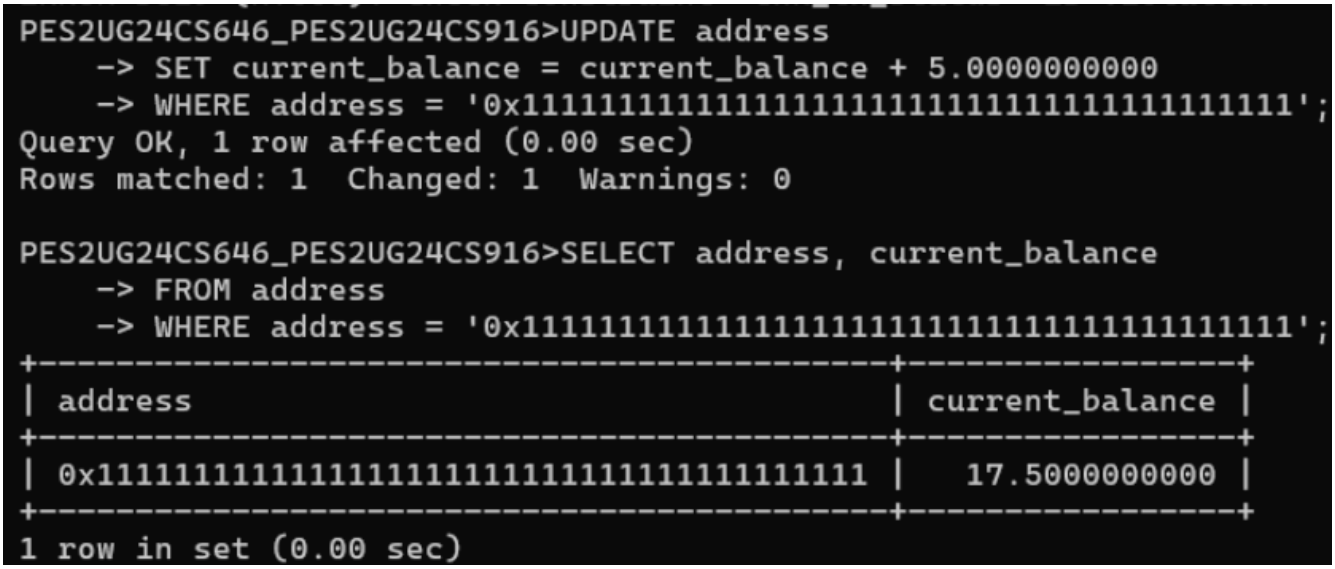
```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     from_address,
    ->     COUNT(*) AS transaction_count
    -> FROM blockchain_transaction
    -> GROUP BY from_address
    -> HAVING COUNT(*) > 1;
+------------------------------------------+-------------------+
| from_address                             | transaction_count |
+------------------------------------------+-------------------+
| 0x1111111111111111111111111111111111111111 |                 2 |
| 0x4444444444444444444444444444444444444444 |                 2 |
| 0x5555555555555555555555555555555555555555 |                 2 |
| 0x6666666666666666666666666666666666666666 |                 2 |
| 0x7777777777777777777777777777777777777777 |                 2 |
| 0x8888888888888888888888888888888888888888 |                 2 |
| 0x9999999999999999999999999999999999999999 |                 2 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |                 2 |
+------------------------------------------+-------------------+
8 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     token_symbol,
    ->     SUM(amount) AS total_amount
    -> FROM token_transfer
    -> GROUP BY token_symbol
    -> HAVING SUM(amount) > 1000;
+--------------+-----------------------+
| token_symbol | total_amount          |
+--------------+-----------------------+
| USDT         | 11000.000000000000000000 |
| USDC         |  2350.000000000000000000 |
| DAI          |  1550.000000000000000000 |
+--------------+-----------------------+
3 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     COUNT(*) AS total_transactions,
    ->     SUM(amount) AS total_amount,
    ->     AVG(amount) AS average_amount,
    ->     MIN(amount) AS minimum_amount,
    ->     MAX(amount) AS maximum_amount
    -> FROM blockchain_transaction;
+--------------------+----------------+-------------------+----------------+----------------+
| total_transactions | total_amount   | average_amount    | minimum_amount | maximum_amount |
+--------------------+----------------+-------------------+----------------+----------------+
|                 20 | 272.5000000000 | 13.62500000000000 |   2.7500000000 |  40.0000000000 |
+--------------------+----------------+-------------------+----------------+----------------+
1 row in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     COUNT(*) AS total_transactions,
    ->     SUM(gas_fee) AS total_gas_fee,
    ->     AVG(gas_fee) AS average_gas_fee,
    ->     MIN(gas_fee) AS minimum_gas_fee,
    ->     MAX(gas_fee) AS maximum_gas_fee
    -> FROM blockchain_transaction;
+--------------------+---------------+------------------+-----------------+-----------------+
| total_transactions | total_gas_fee | average_gas_fee  | minimum_gas_fee | maximum_gas_fee |
+--------------------+---------------+------------------+-----------------+-----------------+
|                 20 |  0.0628000000 | 0.00314000000000 |    0.0012000000 |    0.0060000000 |
+--------------------+---------------+------------------+-----------------+-----------------+
1 row in set (0.00 sec)
```

7.2 UPDATE OPERATION



```
PES2UG24CS646_PES2UG24CS916>UPDATE address
    -> SET current_balance = current_balance + 5.0000000000
    -> WHERE address = '0x1111111111111111111111111111111111111111';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS646_PES2UG24CS916>SELECT address, current_balance
    -> FROM address
    -> WHERE address = '0x1111111111111111111111111111111111111111';
+------------------------------------------+-----------------+
| address                                  | current_balance |
+------------------------------------------+-----------------+
| 0x1111111111111111111111111111111111111111 |   17.5000000000 |
+------------------------------------------+-----------------+
1 row in set (0.00 sec)
```

7.3 DELETE OPERATION

```
PES2UG24CS646_PES2UG24CS916>DELETE FROM watchlist_address
    -> WHERE watchlist_address_id = 15;
Query OK, 1 row affected (0.00 sec)

PES2UG24CS646_PES2UG24CS916>SELECT *
    -> FROM watchlist_address;
+----------------------+---------------------+--------------+--------------------------------------------+
| watchlist_address_id | added_at            | watchlist_id | address                                    |
+----------------------+---------------------+--------------+--------------------------------------------+
|                    1 | 2026-10-03 21:31:33 |            1 | 0x1111111111111111111111111111111111111111 |
|                    2 | 2026-10-03 21:31:33 |            1 | 0x5555555555555555555555555555555555555555 |
|                    3 | 2026-10-03 21:31:33 |            2 | 0x3333333333333333333333333333333333333333 |
|                    4 | 2026-10-03 21:31:33 |            2 | 0x6666666666666666666666666666666666666666 |
|                    5 | 2026-10-03 21:31:33 |            3 | 0x9999999999999999999999999999999999999999 |
|                    6 | 2026-10-03 21:31:33 |            3 | 0xcccccccccccccccccccccccccccccccccccccccc |
|                    7 | 2026-10-03 21:31:33 |            4 | 0x2222222222222222222222222222222222222222 |
|                    8 | 2026-10-03 21:31:33 |            4 | 0x8888888888888888888888888888888888888888 |
|                    9 | 2026-10-03 21:31:33 |            5 | 0x3333333333333333333333333333333333333333 |
|                   10 | 2026-10-03 21:31:33 |            5 | 0x9999999999999999999999999999999999999999 |
|                   11 | 2026-10-03 21:31:33 |            6 | 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |
|                   12 | 2026-10-03 21:31:33 |            7 | 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb |
|                   13 | 2026-10-03 21:31:33 |            8 | 0x7777777777777777777777777777777777777777 |
|                   14 | 2026-10-03 21:31:33 |            9 | 0x8888888888888888888888888888888888888888 |
+----------------------+---------------------+--------------+--------------------------------------------+
14 rows in set (0.00 sec)
```

7.4 CORRELATED QUERY

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     a.address,
    ->     a.current_balance
    -> FROM address a
    -> WHERE EXISTS
    -> (
    ->     SELECT 1
    ->     FROM blockchain_transaction t
    ->     WHERE t.from_address = a.address
    -> );
+--------------------------------------------+-----------------+
| address                                    | current_balance |
+--------------------------------------------+-----------------+
| 0x1111111111111111111111111111111111111111 |   16.0000000000 |
| 0x2222222222222222222222222222222222222222 |   27.2500000000 |
| 0x3333333333333333333333333333333333333333 |  100.0000000000 |
| 0x4444444444444444444444444444444444444444 |    8.2500000000 |
| 0x5555555555555555555555555555555555555555 |   50.0000000000 |
| 0x6666666666666666666666666666666666666666 |   75.5000000000 |
| 0x7777777777777777777777777777777777777777 |   18.7500000000 |
| 0x8888888888888888888888888888888888888888 |   32.0000000000 |
| 0x9999999999999999999999999999999999999999 |  200.0000000000 |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa |    5.5000000000 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb |   45.0000000000 |
| 0xcccccccccccccccccccccccccccccccccccccccc |  150.0000000000 |
+--------------------------------------------+-----------------+
12 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     a.address,
    ->     a.current_balance
    -> FROM address a
    -> WHERE NOT EXISTS
    -> (
    ->     SELECT 1
    ->     FROM blockchain_transaction t
    ->     WHERE t.from_address = a.address
    ->        OR t.to_address = a.address
    -> );
Empty set (0.00 sec)
```

7.5 NESTED QUERY

```
PES2UG24CS646_PES2UG24CS916>SELECT DISTINCT address
    -> FROM address
    -> WHERE address IN
    -> (
    ->     SELECT from_address
    ->     FROM blockchain_transaction
    ->     WHERE amount >
    ->     (
    ->         SELECT AVG(amount)
    ->         FROM blockchain_transaction
    ->     )
    -> );
+--------------------------------------------+
| address                                    |
+--------------------------------------------+
| 0x4444444444444444444444444444444444444444 |
| 0x5555555555555555555555555555555555555555 |
| 0x7777777777777777777777777777777777777777 |
| 0x8888888888888888888888888888888888888888 |
| 0x9999999999999999999999999999999999999999 |
| 0xbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbbb |
+--------------------------------------------+
6 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     address,
    ->     current_balance
    -> FROM address
    -> WHERE current_balance >
    -> (
    ->     SELECT AVG(current_balance)
    ->     FROM address
    -> );
+--------------------------------------------+-----------------+
| address                                    | current_balance |
+--------------------------------------------+-----------------+
| 0x3333333333333333333333333333333333333333 |  100.0000000000 |
| 0x6666666666666666666666666666666666666666 |   75.5000000000 |
| 0x9999999999999999999999999999999999999999 |  200.0000000000 |
| 0xcccccccccccccccccccccccccccccccccccccccc |  150.0000000000 |
+--------------------------------------------+-----------------+
4 rows in set (0.00 sec)
```

```
PES2UG24CS646_PES2UG24CS916>SELECT
    ->     block_number,
    ->     transaction_count
    -> FROM block
    -> WHERE transaction_count >
    -> (
    ->     SELECT AVG(transaction_count)
    ->     FROM block
    -> );
Empty set (0.00 sec)
```

8. STORED PROCEDURES, FUNCTIONS AND TRIGGERS

8.1 STORED PROCEDURES OR FUNCTIONS

```
PES2UG24CS646_PES2UG24CS916>DELIMITER //
PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>CREATE PROCEDURE GetBlockTransactions(IN p_block_number BIGINT)
    -> BEGIN
    ->     SELECT
    ->         tx_hash,
    ->         amount,
    ->         gas_fee,
    ->         status,
    ->         timestamp,
    ->         from_address,
    ->         to_address
    ->     FROM blockchain_transaction
    ->     WHERE block_number = p_block_number;
    -> END //
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>DELIMITER ;
PES2UG24CS646_PES2UG24CS916>CALL GetBlockTransactions(100001);
+------------------------------------------------------------------+------------+----------+---------+---------------------+------------------------------------------------------------------+------------------------------------------------------------------+
| tx_hash                                                          | amount     | gas_fee  | status  | timestamp           | from_address                                                     | to_address                                                       |
+------------------------------------------------------------------+------------+----------+---------+---------------------+------------------------------------------------------------------+------------------------------------------------------------------+
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | 5.50000000 | 0.002100 | Success | 2026-09-01 10:05:00 | 0x1111111111111111111111111111111111111111                       | 0x2222222222222222222222222222222222222222                       |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaab | 8.25000000 | 0.002500 | Success | 2026-09-01 10:08:00 | 0x2222222222222222222222222222222222222222                       | 0x3333333333333333333333333333333333333333                       |
+------------------------------------------------------------------+------------+----------+---------+---------------------+------------------------------------------------------------------+------------------------------------------------------------------+
2 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

```
PES2UG24CS646_PES2UG24CS916>DELIMITER //
PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>CREATE PROCEDURE GetAddressTransactions(IN p_address VARCHAR(42))
    -> BEGIN
    ->     SELECT
    ->         tx_hash,
    ->         amount,
    ->         gas_fee,
    ->         status,
    ->         timestamp,
    ->         block_number,
    ->         from_address,
    ->         to_address
    ->     FROM blockchain_transaction
    ->     WHERE from_address = p_address
    ->        OR to_address = p_address
    ->     ORDER BY timestamp DESC;
    -> END //
Query OK, 0 rows affected (0.00 sec)

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>DELIMITER ;
PES2UG24CS646_PES2UG24CS916>CALL GetAddressTransactions(
    -> '0x1111111111111111111111111111111111111111'
    -> );
+------------------------------------------------------------------+------------+----------+---------+---------------------+--------------+------------------------------------------------------------------+------------------------------------------------------------------+
| tx_hash                                                          | amount     | gas_fee  | status  | timestamp           | block_number | from_address                                                     | to_address                                                       |
+------------------------------------------------------------------+------------+----------+---------+---------------------+--------------+------------------------------------------------------------------+------------------------------------------------------------------+
| 0x1234567890abcdef1234567890abcdef1234567890abcdef1234567890abcdef | 3.50000000 | 0.001200 | Success | 2026-09-07 16:00:00 |       100007 | 0x1111111111111111111111111111111111111111                       | 0x4444444444444444444444444444444444444444                       |
| 0xfffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffe | 4.25000000 | 0.001700 | Success | 2026-09-06 15:55:00 |       100006 | 0xcccccccccccccccccccccccccccccccccccccccc                       | 0x1111111111111111111111111111111111111111                       |
| 0xaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa | 5.50000000 | 0.002100 | Success | 2026-09-01 10:05:00 |       100001 | 0x1111111111111111111111111111111111111111                       | 0x2222222222222222222222222222222222222222                       |
+------------------------------------------------------------------+------------+----------+---------+---------------------+--------------+------------------------------------------------------------------+------------------------------------------------------------------+
3 rows in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)
```

The project implements two stored procedures:

1. GetBlockTransactions: Accepts a block number as input and retrieves all blockchain transactions associated with that block.
2. GetAddressTransactions: Accepts a wallet address as input and retrieves all transactions in which the specified address acts as either the sender or receiver. The results are sorted by timestamp in descending order.

These procedures simplify data retrieval and allow frequently used database operations to be executed efficiently.

8.2 TRIGGERS

```
PES2UG24CS646_PES2UG24CS916>DELIMITER //
PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>CREATE TRIGGER validate_transaction_amount
    -> BEFORE INSERT ON blockchain_transaction
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.amount <= 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Transaction amount must be greater than zero';
    ->     END IF;
    ->
    ->     IF NEW.gas_fee < 0 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Gas fee cannot be negative';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS646_PES2UG24CS916>
PES2UG24CS646_PES2UG24CS916>DELIMITER ;
PES2UG24CS646_PES2UG24CS916>INSERT INTO blockchain_transaction
    -> (tx_hash, amount, gas_fee, status, timestamp,
    ->  block_number, from_address, to_address)
    -> VALUES
    -> (
    -> '0x8888888888888888888888888888888888888888888888888888888888888888',
    -> 0,
    -> 0.0020000000,
    -> 'Success',
    -> '2026-10-03 21:00:00',
    -> 100001,
    -> '0x1111111111111111111111111111111111111111',
    -> '0x2222222222222222222222222222222222222222'
    -> );
ERROR 1644 (45000): Transaction amount must be greater than zero
```

The project implements a BEFORE INSERT trigger named validate_transaction_amount on the blockchain_transaction table.

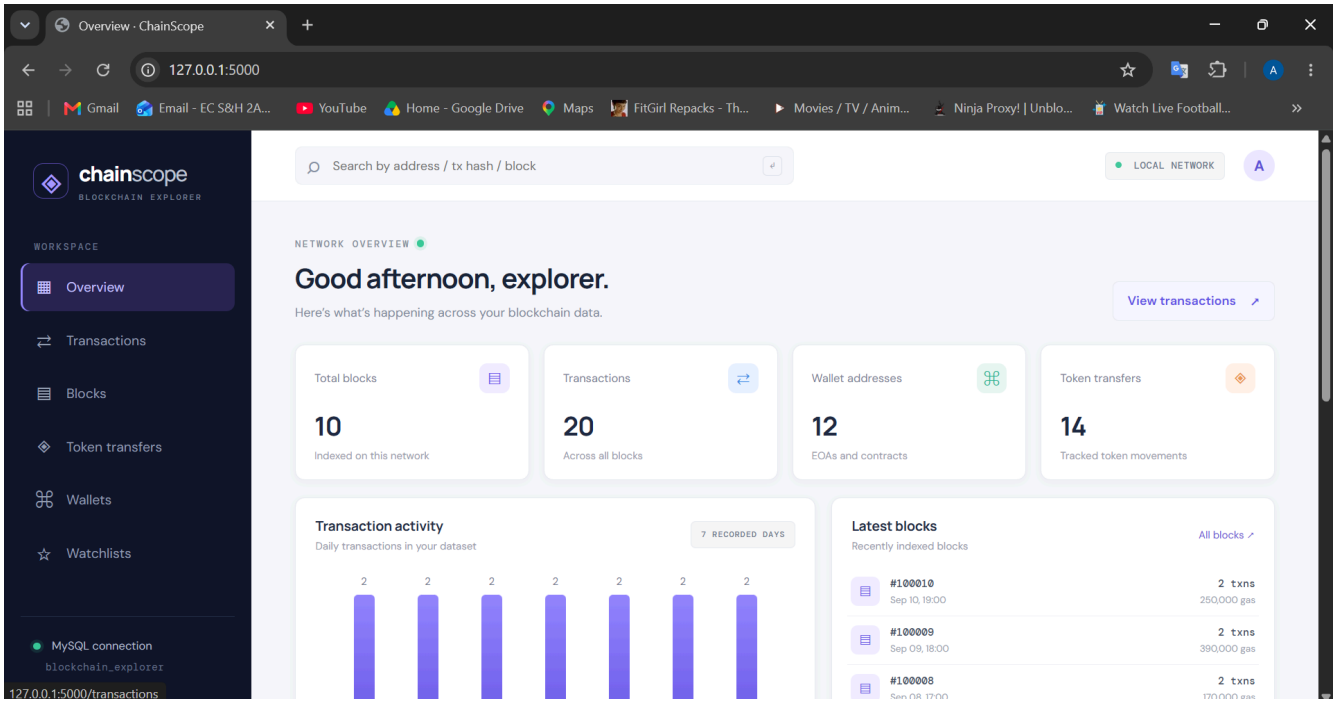
The trigger performs the following validations:

1. Checks whether the transaction amount is greater than zero.
2. Ensures that the gas fee is not negative.
3. Generates an error using SIGNAL SQLSTATE '45000' if either condition is violated.

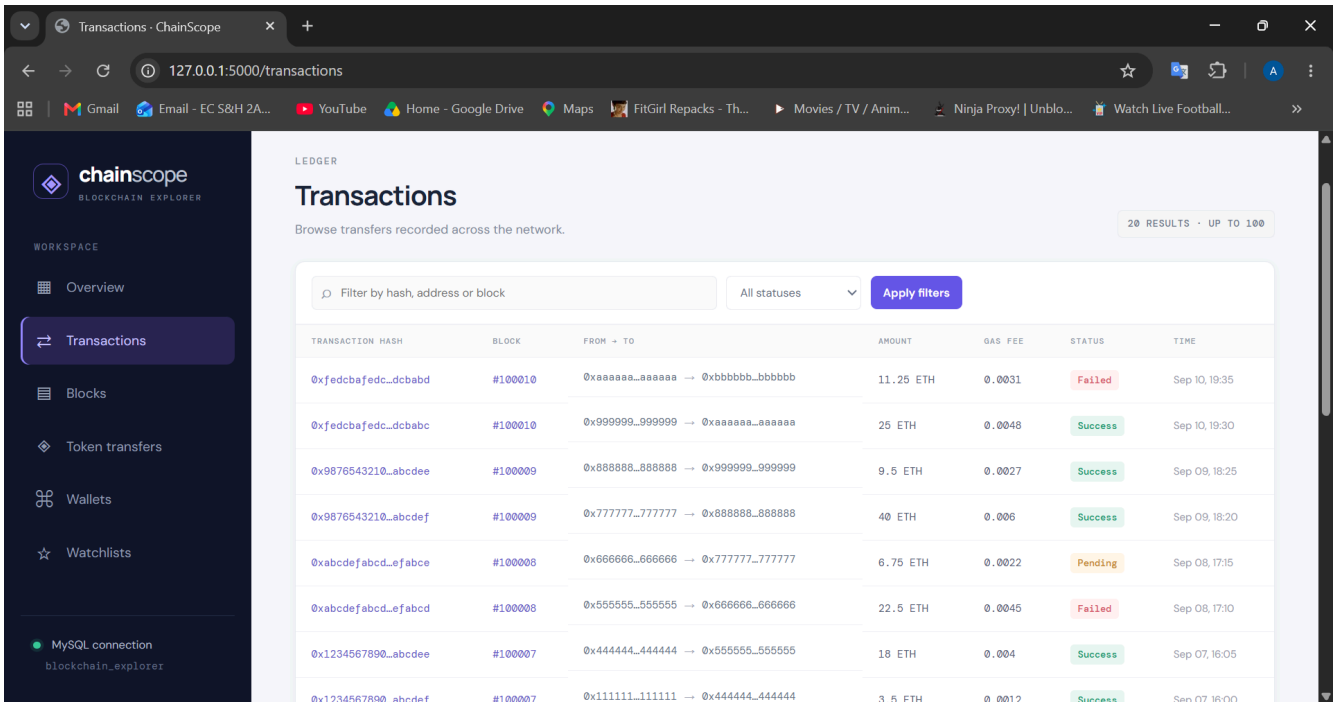
This trigger helps maintain data integrity by preventing invalid transaction records from being inserted into the database.

9. FRONT END DEVELOPEMENT

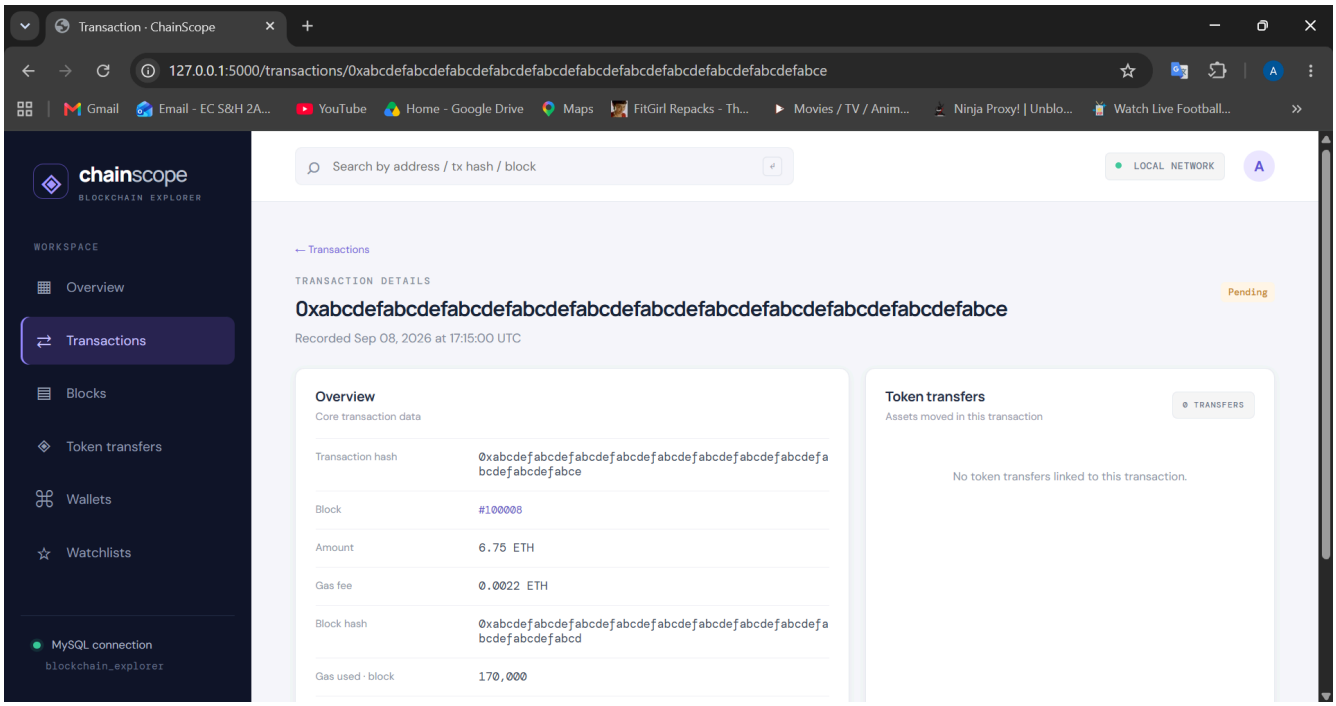
9.1 Home Page / Dashboard



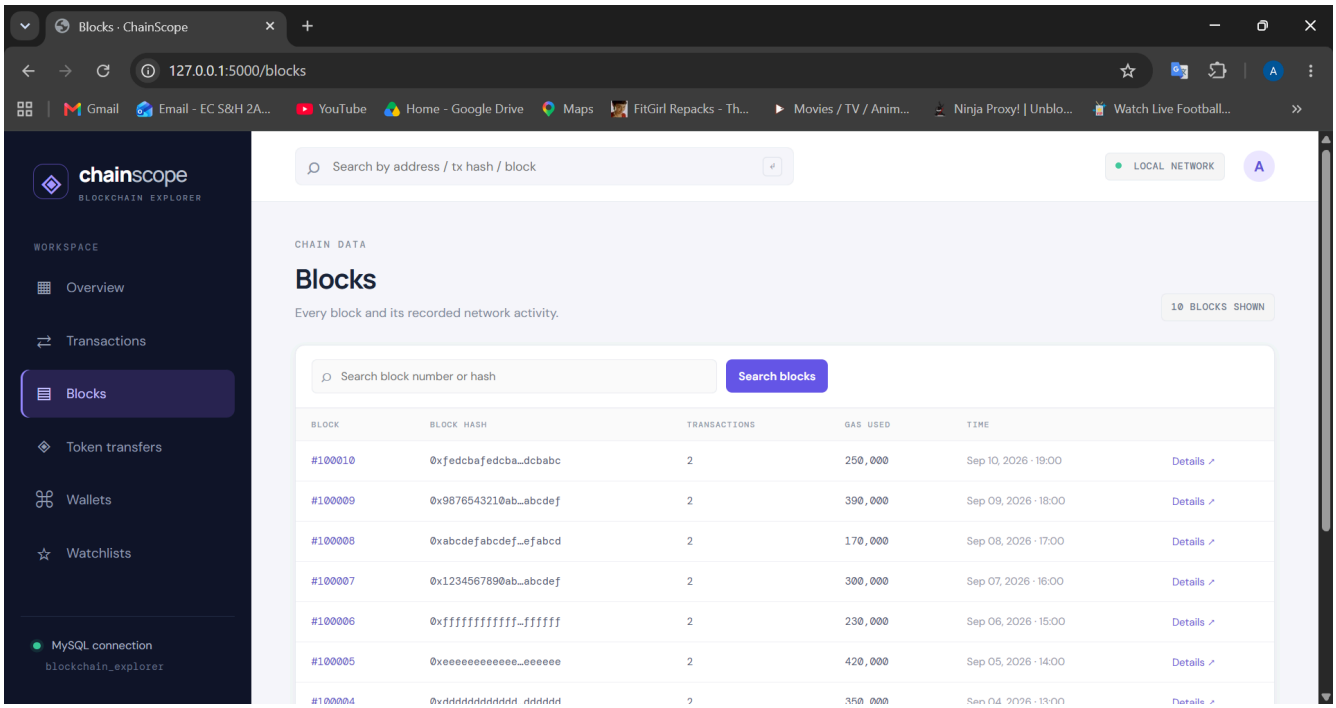
9.2 Transaction Explorer



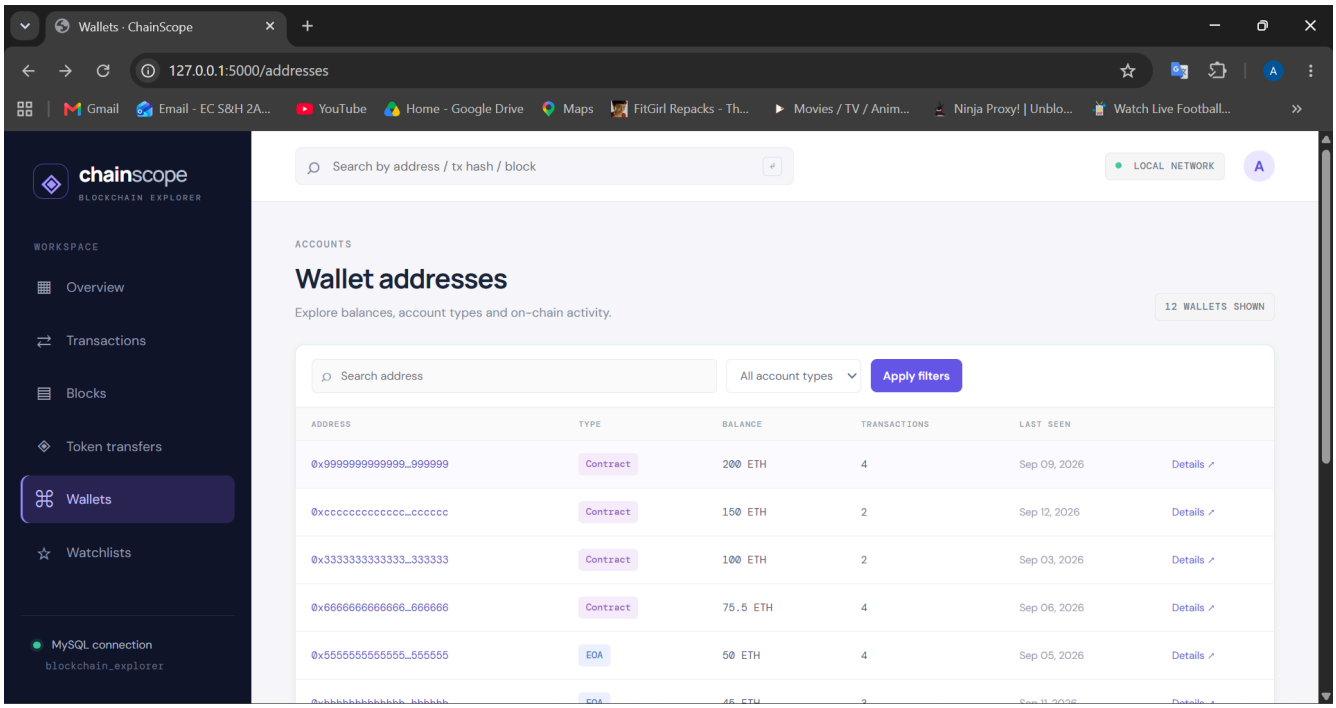
9.3 Transaction Details



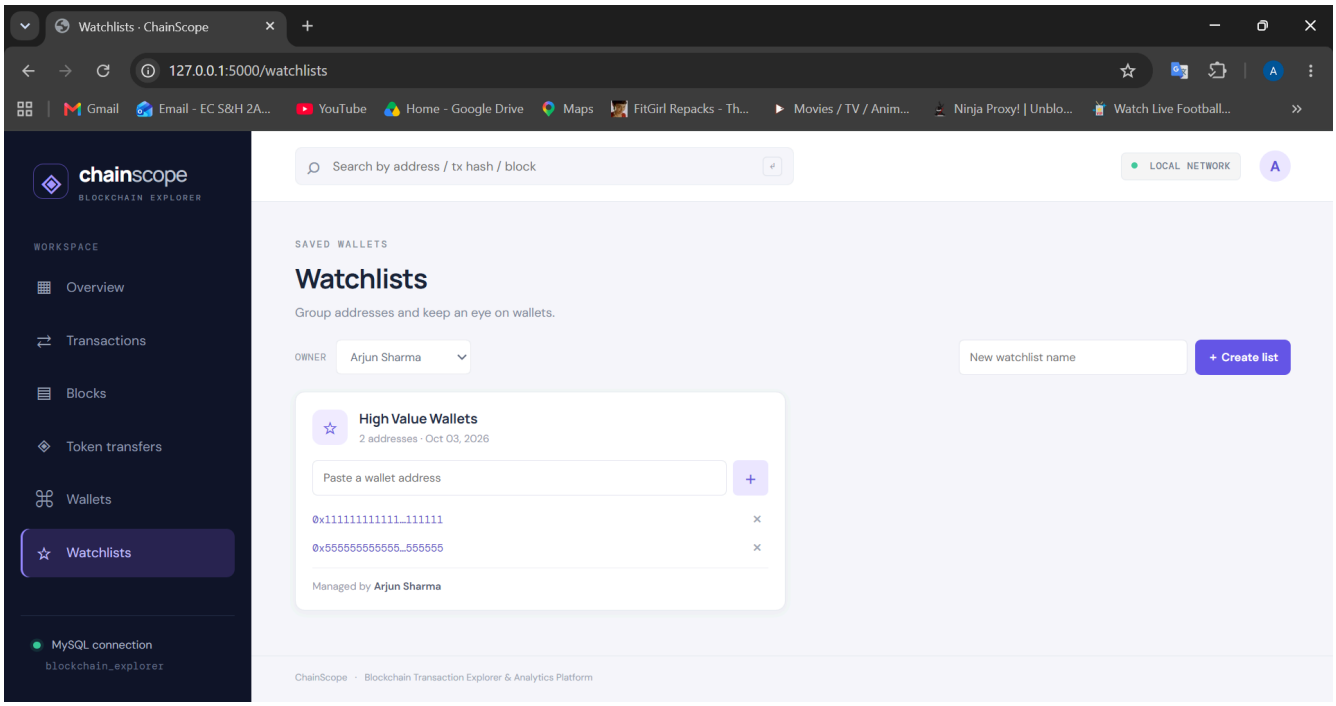
9.4 Block Explorer



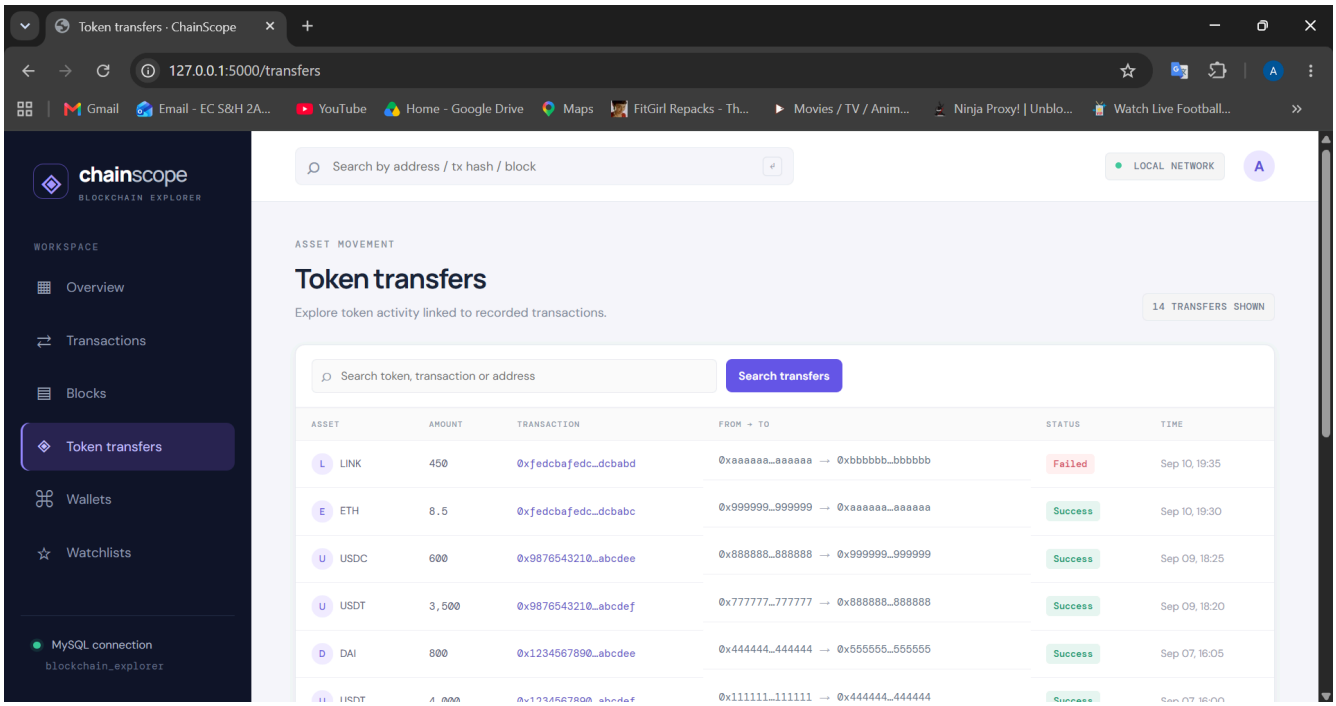
9.5 Wallet Address Explorer



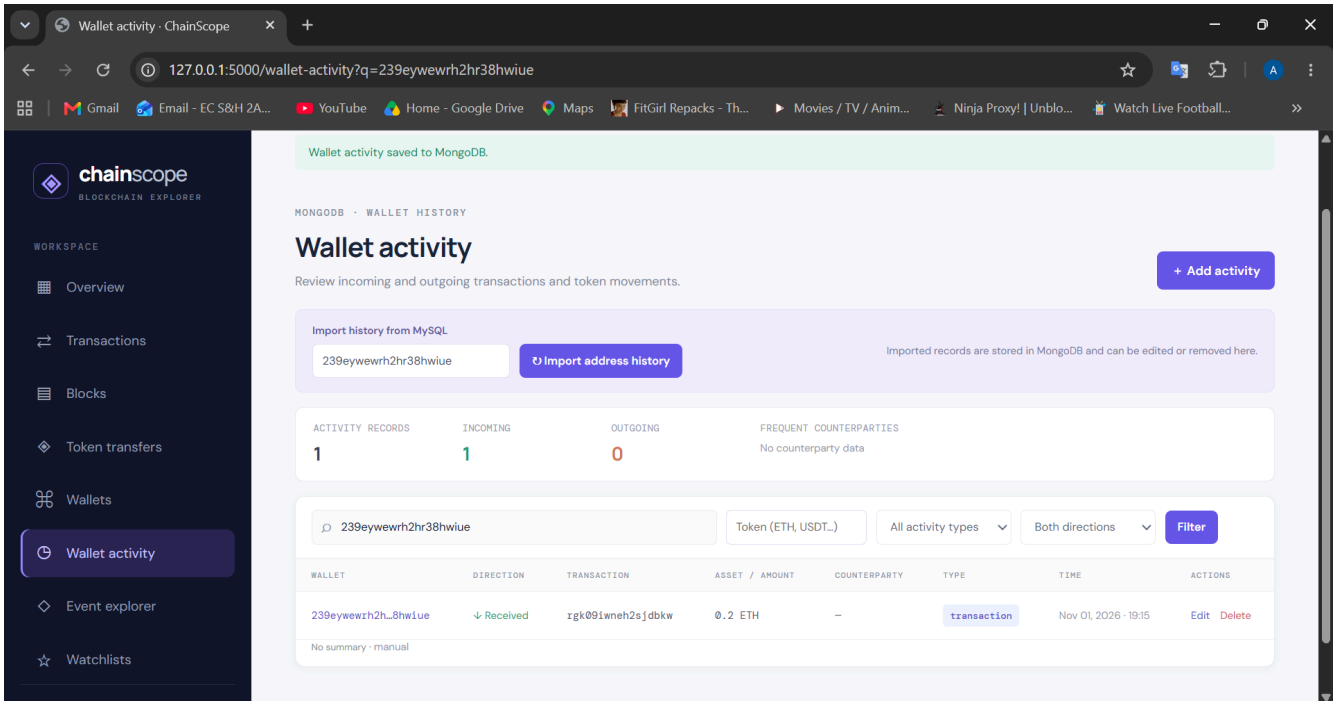
9.6 Watchlist Management



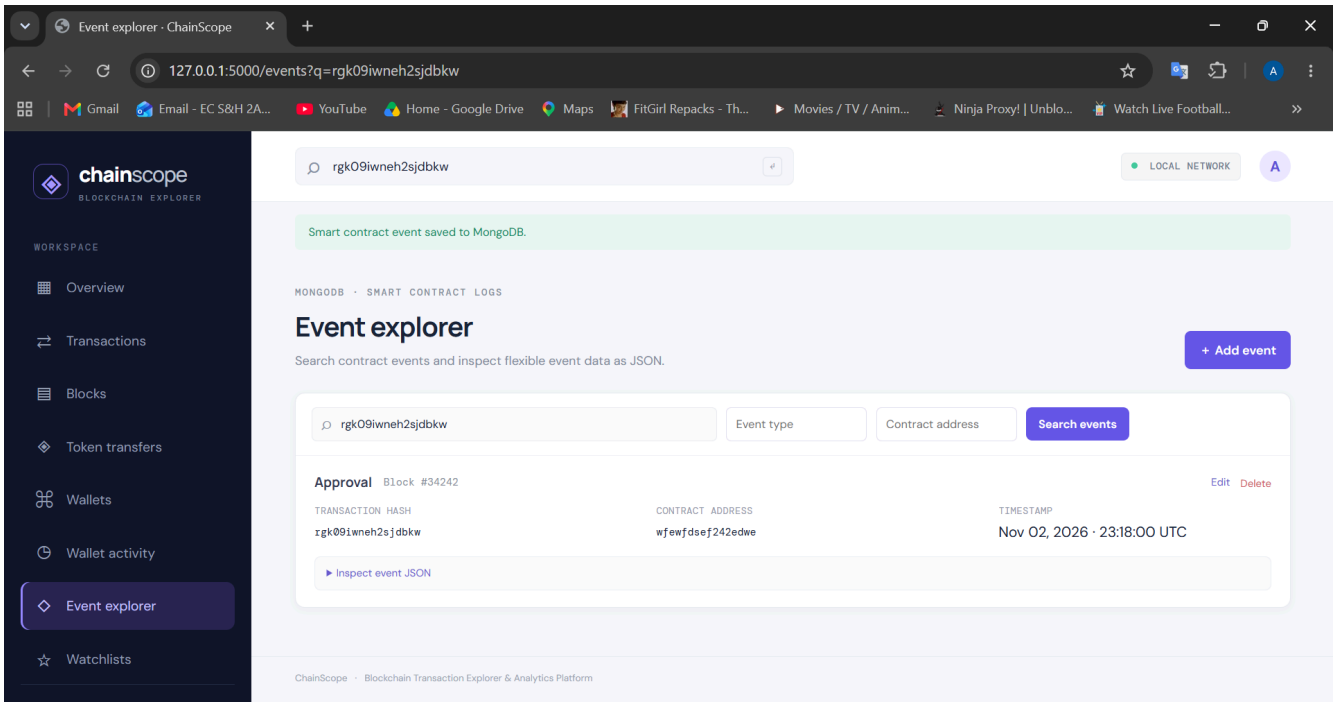
9.7 Token Transfer Explorer



9.8 Wallet Activity (NoSQL)



9.9 Event Explorer (NoSQL)



REFERENCES

1. MySQL Documentation – https://dev.mysql.com/doc/
2. MySQL 8.0 Reference Manual – https://dev.mysql.com/doc/refman/8.0/en/
3. Ethereum Documentation – https://ethereum.org/en/developers/docs/
4. Database System Concepts – Abraham Silberschatz, Henry F. Korth and S. Sudarshan.
5. PES University DBMS course materials and laboratory resources.

GitHub Link:

https://github.com/AbhiramKolalapudi/blockchain-transaction-explorer

Appendix A: Definitions, Acronyms and Abbreviations

Term / Acronym	Definition
DBMS	Database Management System
SQL	Structured Query Language
NoSQL	Non-relational database approach
RDBMS	Relational Database Management System
PK	Primary Key
FK	Foreign Key
DDL	Data Definition Language
DML	Data Manipulation Language
TCL	Transaction Control Language
MongoDB	Document-oriented NoSQL database
ER Model	Entity-Relationship Model
MySQL	Relational Database Management System
Constraint	Rule used to maintain data integrity
Trigger	Automatically executed SQL operation
Stored Procedure	Stored collection of SQL statements
View	Virtual table based on an SQL query
Transaction	Group of database operations treated as one unit
Blockchain	Distributed ledger that stores transactions in blocks
TX Hash	Unique identifier of a blockchain transaction
EOA	Externally Owned Account
Gas Fee	Fee paid to process a blockchain transaction
Token Transfer	Movement of tokens between blockchain addresses